

A Period-Three Limit Orbit in a Thue–Morse Heteroclinic Closure

Route-A structural certificate C154

Abstract

We join a period-three past to a Thue–Morse future in one two-sided symbolic configuration and close its full shift orbit. The closure has exactly the Thue–Morse component, the interface orbit, and the three periodic phases. Consequently its only primitive orbit has length three and its Artin–Mazur zeta is $(1 - z^3)^{-1}$. This periodic layer is generated by one heteroclinic closure rather than freely attached. The full two-sided orbit is dense, but isolated wandering interface points make standard forward topological transitivity fail.

Keywords: symbolic dynamics; Thue–Morse subshift; heteroclinic orbit closure; wandering point; nonwandering set; Artin–Mazur zeta.

中文摘要

本文考察一个双侧符号构型：左半轴取三周期背景 234，右半轴取 Thue–Morse 序列，并对其完整移位轨道取闭包。我们证明该闭包恰由 Thue–Morse 子移位、接口轨道以及三周期轨道组成；接口点均为孤立游荡点，非游荡集正好是两个极限分量。所有周期点仅来自三周期分量，因此固定点计数在 $3 \mid n$ 时为 3，其余为零，Artin–Mazur ζ 函数为 $(1 - z^3)^{-1}$ 。周期骨架由单个异宿轨道闭包内生，而非自由拼接。需要强调，稠密的是完整双侧 $\mathbb{Z}$ 轨道；标准前向拓扑传递性反而失败。

关键词：符号动力系统；Thue–Morse 子移位；异宿轨道闭包；游荡点；非游荡集；Artin–Mazur ζ 函数。

1 The frozen interface and its closure

Let t_n be binary digit-sum parity for $n \geq 0$, and let $y_j = (2, 3, 4)_{j \bmod 3}$. Define

$$x_j = \begin{cases} y_j, & j < 0, \\ t_j, & j \geq 0, \end{cases} \quad X = \overline{\{\sigma^n x : n \in \mathbb{Z}\}},$$

where $(\sigma x)_j = x_{j+1}$. If $n_k \rightarrow +\infty$, every fixed window of $\sigma^{n_k} x$ eventually lies in the binary tail, so each limit lies in X_{TM} . Conversely, uniform recurrence puts every finite word of every X_{TM} point arbitrarily far out in t ; nested occurrences and a diagonal subsequence realize every such point as a positive-shift limit. If $n_k \rightarrow -\infty$, every fixed window lies eventually in the periodic tail; one residue class modulo three gives one phase of y , and all phases occur. A bounded convergent integer-shift sequence is eventually constant, while an unbounded sequence has a subsequence escaping in one direction. Thus

$$X = X_{\text{TM}} \sqcup \text{Orb}(x) \sqcup \text{Orb}(y). \quad (1)$$

The pair 40 occurs exactly once in $\sigma^n x$, at coordinates $-n-1, -n$, and occurs in neither limit component. Its cylinder is the singleton $\{\sigma^n x\}$, so every interface point is isolated and wandering. The full two-sided $\mathbb{Z}$ -orbit is dense by construction. This is not standard forward topological transitivity: the open sets $U = \{\sigma x\}$ and $V = \{x\}$ satisfy $\sigma^n(U) \cap V = \emptyset$ for every $n \geq 0$. The period-three orbit is a proper closed invariant set, so X is not minimal.

2 Nonwandering and periodic sets

The Thue–Morse language has no periodic point. For a proposed period p , choose odd k above its binary length and put $d = p(2^k - 1)$. Then $p \mid d$ and $\text{popcount}(d) = k$, so $t_d \neq t_0$. If b is the binary length of d , every length- 2^{b+1} interval contains a complete 2^b -aligned block; its symbols at offsets $0, d$ are opposite. Hence no language window of that length is p -periodic.

The isolated wandering interface orbit has no periodic point. Points of the minimal Thue–Morse component are nonwandering, as are the three phases of y ; (1) exhausts the complement. Therefore

$$\Omega(\sigma) = X_{\text{TM}} \sqcup \text{Orb}(y), \quad \# \text{Fix}(\sigma^n) = \begin{cases} 3, & 3 \mid n, \\ 0, & 3 \nmid n. \end{cases} \quad (2)$$

Möbius inversion leaves three exact-period points at $n = 3$, hence one primitive cycle. Moreover

$$\log \zeta_X(z) = \sum_{q \geq 1} \frac{3z^{3q}}{3q} = -\log(1 - z^3), \quad \zeta_X(z) = \frac{1}{1 - z^3}. \quad (3)$$

The exact ledgers are sentinels, not theorem cutoffs. An independent checker passes 549 assertions, SymPy passes 357 checks, replay is byte-identical, and 48 hostile cases are rejected. The strict tuple is

(A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).

We claim no target divisor, arithmetic/local factor, target functional equation or counting law, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope:

NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data and code availability. The package-local `results/c154_heteroclinic_evidence.json` and `code/` directory contain the exact certificate, deterministic producer, independent checker, symbolic cross-check, replay, and mutation audit.

Ethics. This mathematical and symbolic-computational study uses no human participants, animals, personal data, clinical records, or sensitive data; research ethics approval is not applicable.

Author contributions (CRediT). This anonymous technical certificate does not assign individual authorship. Package provenance records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing at the artifact level. AI systems are not authors.

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AI-use disclosure. An AI coding assistant supported drafting, exact-code development, and internal proof checking. It was not an external reviewer. Released claims are exposed to deterministic reproduction, independent checking, symbolic cross-checking, and hostile mutation tests in the package.